

Affine-DDPNM: Methodology

Version 6 — revised 2026-08-07

This revision fixes the traction/load sign convention, the orientation of tangential modes, the assumptions for local invertibility and global SPD, and the interpretation of the Galerkin enrichment statement.

We extend the domain-decomposition pore-network method (DD-PNM) of [1] by enriching the interface traction space from a single constant-normal mode to a nine-mode affine representation.

Scope. The formulation and the accompanying Galerkin analysis are stated for **planar interfaces** with a fixed orthonormal frame and dimensionless coordinates (s, t) . The watershed implementation on non-planar, multi-facet interfaces is a numerical extension and is *not* covered by the present approximation theorem.

1 Model problem and weak-form convention

Steady incompressible Stokes flow in $\Omega \subset \mathbb{R}^3$:

$$-\nu \Delta \mathbf{u} + \nabla p = \mathbf{0}, \quad \text{in } \Omega, \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0, \quad \text{in } \Omega, \quad (2)$$

$$\mathbf{u} = \mathbf{0}, \quad \text{on } \Gamma_{\text{wall}}, \quad (3)$$

$$\boldsymbol{\sigma}(\mathbf{u}, p) \mathbf{n} = -p_b \mathbf{n}, \quad \text{on } \Gamma_{\text{in}} \cup \Gamma_{\text{out}}, \quad (4)$$

where $\boldsymbol{\sigma}(\mathbf{u}, p) := -pI + \nu \nabla \mathbf{u}$ is the pseudostress. The inlet/outlet condition prescribes the normal pseudotraction, not the pressure directly: $\mathbf{n} \cdot \boldsymbol{\sigma} \mathbf{n} = -p_b$ does not imply $p = p_b$ point-wise because the normal viscous term contributes. The incompressible weak formulation uses $b_i(\mathbf{v}, q) = -\int_{\Omega_i} q \nabla \cdot \mathbf{v} \, dx$, giving the local discrete system

$$\begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} U_i \\ P_i \end{bmatrix} = \begin{bmatrix} f_i \\ \mathbf{0} \end{bmatrix}. \quad (5)$$

No pressure stabilisation is used.

2 Domain partition and oriented interface frames

Ω is partitioned into N non-overlapping subdomains $\{\Omega_i\}_{i=1}^N$. Each internal interface γ_k is planar and shared by pores (i_k, j_k) . Choose once a global orthonormal frame $\{\mathbf{n}_k, \mathbf{t}_{k,1}, \mathbf{t}_{k,2}\}$ with $\mathbf{n}_k$ directed from i_k to j_k . Let $\sigma_{i,k} = +1$ for $i = i_k$ and -1 for $i = j_k$.

To match the code's assembled normal direction $-\mathbf{n}_{i,k}$ and tangential directions $\sigma_{i,k} \mathbf{t}_{k,\nu}$, define **oriented inward-traction directions**

$$\mathbf{d}_{i,k,n} = \mathbf{n}_{i,k}, \quad \mathbf{d}_{i,k,t_\nu} = -\sigma_{i,k} \mathbf{t}_{k,\nu} \quad (\nu = 1, 2). \quad (6)$$

The minus sign in the tangential definition is an orientation choice; changing it together with the corresponding coefficient changes no traction space. Crucially,

$$\mathbf{d}_{j,k,\beta} = -\mathbf{d}_{i,k,\beta} \quad (\beta = n, t_1, t_2). \quad (7)$$

Half-span-normalised in-plane coordinates $s_k(\mathbf{x}) = ((\mathbf{x} - \mathbf{c}_k) \cdot \mathbf{t}_{k,1})/a_k$, $t_k(\mathbf{x}) = ((\mathbf{x} - \mathbf{c}_k) \cdot \mathbf{t}_{k,2})/b_k$, with scalar shapes $\varphi_0 = 1$, $\varphi_s = s_k$, $\varphi_t = t_k$, $\mathcal{M} = \{0, s, t\} \times \{n, t_1, t_2\}$.

Column ordering. The *code* uses component-major ordering

$$\begin{aligned} \mathcal{O}_{\text{code}} = & ((0, n), (s, n), (t, n), \\ & (0, t_1), (s, t_1), (t, t_1), \\ & (0, t_2), (s, t_2), (t, t_2)). \end{aligned}$$

The companion mathematical-properties document uses polynomial-major ordering

$$\begin{aligned} \mathcal{O}_{\text{math}} = & ((0, n), (0, t_1), (0, t_2), \\ & (s, n), (s, t_1), (s, t_2), \\ & (t, n), (t, t_1), (t, t_2)). \end{aligned}$$

The zero-based permutation between the two orderings is

$$p = (0, 3, 6, 1, 4, 7, 2, 5, 8).$$

Let Π be the corresponding 9×9 permutation matrix; then $\boldsymbol{\lambda}_k^{\text{math}} = \Pi \boldsymbol{\lambda}_k^{\text{code}}$ and the per-interface compliance block transforms as $\mathbf{G}_k^{\text{math}} = \Pi \mathbf{G}_k^{\text{code}} \Pi^\top$. Note that $\Pi = \Pi^\top = \Pi^{-1}$ (this particular permutation is self-inverse). The coefficient vector $\boldsymbol{\lambda}_k \in \mathbb{R}^9$ follows the code ordering throughout this document; the mathematical properties document uses the polynomial-major convention.

Terminology. The variable $-\sigma \mathbf{n}$ appearing in the traction Ansatz is the *negative outward traction*, or equivalently the inward-acting pressure-oriented traction. We refer to it as “inward traction” for brevity; it is not a distinct physical quantity but a sign-flipped pseudotraction.

3 Affine traction Ansatz and code-compatible load sign

Assumption 1 (Local algebraic stability). *For every pore i :*

- (a) A_i is SPD.
- (b) B_i has full row rank (discrete inf-sup);
- (c) the constant function 1 belongs to the local pressure space.

For the standard conforming velocity space, condition (a) follows, for example, if Ω_i is connected and Lipschitz and its essential no-slip boundary portion has positive surface measure. The Poincaré-Friedrichs inequality then excludes nonzero constant-velocity null modes.

The unknown coefficients parameterize the **inward** traction:

$$-\sigma(\mathbf{u}_i, p_i) \mathbf{n}_{i,k} = \sum_{(\alpha, \beta) \in \mathcal{M}} \lambda_{k,\alpha,\beta} \varphi_\alpha \mathbf{d}_{i,k,\beta}. \quad (8)$$

The physical outward traction in the weak form is therefore $\sigma \mathbf{n}_{i,k} = -\sum \lambda \varphi \mathbf{d}_{i,k,\beta}$.

The primitive load vector for mode (α, β) is

$$[\mathbf{f}_{i,k}^{(\alpha,\beta)}]_j = - \int_{\gamma_k} \varphi_\alpha \mathbf{d}_{i,k,\beta} \cdot \boldsymbol{\phi}_j^{(i)} \, dS. \quad (9)$$

The leading minus sign is essential: it converts the inward-traction Ansatz into the physical outward-traction boundary integral. For the normal mode this gives the assembled direction $-\mathbf{n}_{i,k}$; for the tangential modes (6) gives $+\sigma_{i,k} \mathbf{t}_{k,\nu}$, exactly the convention used in `affine_face_basis.py`.

At an inlet/outlet port only the constant normal mode is retained with prescribed coefficient $\lambda_{i,\gamma}^k = p_b$.

4 Local response and compliance map

For each primitive mode solve

$$\begin{bmatrix} A_i & B_i^\top \\ B_i & 0 \end{bmatrix} \begin{bmatrix} U_{i,k}^{(\alpha,\beta)} \\ X_{i,k}^{(\alpha,\beta)} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_{i,k}^{(\alpha,\beta)} \\ \mathbf{0} \end{bmatrix}. \quad (10)$$

Let $n_{u,i}$ denote the number of local velocity DOFs in Ω_i . Let $L_i \in \mathbb{R}^{n_{u,i} \times n_i}$ collect all $n_i = 9m_i^u + m_i^k$ primitive load columns. Define

$$M_i = B_i A_i^{-1} B_i^\top, \quad H_i = A_i^{-1} - A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1}. \quad (11)$$

Then $U_i = H_i L_i \boldsymbol{\lambda}_i$ and the local compliance matrix is

$$\mathbf{G}_i = L_i^\top H_i L_i \in \mathbb{R}^{n_i \times n_i}, \quad (12)$$

symmetric positive semidefinite, with $\boldsymbol{\lambda}_i^\top \mathbf{G}_i \boldsymbol{\lambda}_i = U_i^\top A_i U_i \geq 0$.

For mode (α, β) on interface γ_k , the **weighted velocity moment** is

$$W_{i,k}^{(\alpha,\beta)}(U_i) = \int_{\gamma_k} \varphi_\alpha U_{i,h} \cdot \mathbf{d}_{i,k,\beta} dS. \quad (13)$$

Only $W_{i,k}^{(0,n)}$ is the volumetric flow rate; the other eight are weighted velocity moments (linear normal flux moments and tangential velocity moments). From the load convention, $[\mathbf{G}_i \boldsymbol{\lambda}_i]_{k,\alpha,\beta} = -W_{i,k}^{(\alpha,\beta)}(U_i)$. $\mathbf{G}_i$ is partitioned into $\mathbf{G}_i^{uu}$ (internal–internal) and $\mathbf{G}_i^{uk}$ (internal–boundary).

5 Global Schur complement assembly

Continuity of all nine weighted velocity moments, $W_{i,k}^{(\alpha,\beta)} + W_{j,k}^{(\alpha,\beta)} = 0$, together with Boolean assembly gives

$$\mathbf{S} \boldsymbol{\lambda} = \mathbf{F}, \quad \mathbf{S} = \sum_{i=1}^N (R_i^u)^\top \mathbf{G}_i^{uu} R_i^u, \quad \mathbf{F} = - \sum_{i=1}^N (R_i^u)^\top \mathbf{G}_i^{uk} R_i^k \boldsymbol{\lambda}^k. \quad (14)$$

$\mathbf{S} \in \mathbb{R}^{9m \times 9m}$ is symmetric by construction.

Remark 1 (Conditions for SPD). $\mathbf{S}$ is positive definite under the following conditional hypotheses (see the companion mathematical properties document for full statements and proofs):

1. **Local stability:** Assumption 1.
2. **Local kernel:** $\text{rank}(H_i L_i) = n_i - 1$ for every pore. This condition cannot be reduced to per-face linear independence of the nine modes; it additionally excludes cross-interface dependencies and dependencies introduced by the incompressible projection H_i .
3. **Graph anchoring:** every connected component of the pore adjacency graph contains at least one inlet/outlet boundary port.

Floating basins (pores without a solid-wall facet) violate Assumption 1(a) and are merged into neighbours before assembly. This restores local coercivity but does not supply condition (3); the two issues are logically distinct.

Constant-normal indicator vector. For each pore i , define $\mathbf{1}_{i,0n} \in \mathbb{R}^{n_i}$ by

$$[\mathbf{1}_{i,0n}]_\mu = \begin{cases} 1, & \text{if mode } \mu \text{ is the constant-normal mode } (0, n) \text{ on some port of } \Omega_i, \\ 0, & \text{otherwise.} \end{cases}$$

This vector has entry 1 on every port's $(0, n)$ slot (internal and boundary alike) and 0 on all other slots. In code, the diagnostic mask must be constructed as this specific vector; using `np.ones( $n_i$ )` would produce an incorrect all-ones vector that does *not* represent the constant-pressure kernel.

6 Algorithm

Algorithm 1: Affine-DDPNM solver

Input: Subdomain meshes $\{\Omega_i\}$, FE pairs $(\mathbf{V}_{i,h}, Q_{i,h})$, prescribed boundary coefficients λ^k .

Output: Unknown interface coefficients $\lambda \in \mathbb{R}^{9m}$, local fields $\{(U_i, P_i)\}$, weighted velocity moments, and residuals.

Offline library — for each pore $i = 1, \dots, N$ (in parallel):

1. Assemble A_i, B_i .
2. For each internal interface $\gamma \in \mathcal{U}_i$ and each mode $(\alpha, \beta) \in \mathcal{M}$: assemble $\mathbf{f}_{i,\gamma}^{(\alpha,\beta)}$ via (9), solve (10) for $(U_{i,\gamma}^{(\alpha,\beta)}, X_{i,\gamma}^{(\alpha,\beta)})$.
3. For each boundary port $\gamma \in \mathcal{K}_i$: assemble and solve the single $(0, n)$ mode.
4. Form L_i , compute $\mathbf{G}_i = L_i^\top H_i L_i$ via (12). As a diagnostic, compute the singular values of $H_i L_i$ and confirm one numerically zero singular value (within a stated tolerance) with its right singular vector aligned with $\mathbf{1}_{i,0n}$; equivalently, inspect the eigenvalues of $\mathbf{G}_i$ and confirm one numerically zero eigenvalue with the corresponding eigenvector aligned with $\mathbf{1}_{i,0n}$. (This is the local-kernel condition in Remark 1, item 2. **It is an assumption, not a theorem:** it is not proved for arbitrary pore geometries or meshes and must be verified numerically for each local library.)
5. Accumulate global Schur system from the raw assembled blocks (before any explicit symmetrization): $\mathbf{S} += (R_i^u)^\top \mathbf{G}_i^{uu} R_i^u$, $\mathbf{F} -= (R_i^u)^\top \mathbf{G}_i^{uk} R_i^k \lambda^k$.

Global solve: Check the symmetry defect $\|\mathbf{S} - \mathbf{S}^\top\| / \|\mathbf{S}\|$ of the raw assembled matrix; only then apply $\mathbf{S} \leftarrow \frac{1}{2}(\mathbf{S} + \mathbf{S}^\top)$ if needed. Verify positive definiteness (Cholesky factorization or smallest eigenvalue). Solve $\mathbf{S} \lambda = \mathbf{F}$ by sparse direct solver.

Reconstruction — for each pore $i = 1, \dots, N$ (in parallel):

$$U_i = \sum_{\gamma \in \mathcal{U}_i} \sum_{\alpha, \beta} \lambda_{i,\gamma,\alpha,\beta} U_{i,\gamma}^{(\alpha,\beta)} + \sum_{\gamma \in \mathcal{K}_i} \lambda_{i,\gamma}^k U_{i,\gamma}^{(0,n)},$$

and analogously for P_i with the pressure responses $X_{i,\gamma}^{(\cdot)}$.

Verification (compute independently):

$$\begin{aligned} r_{B,i} &= \|B_i U_i\| / (\|U_i\| + \epsilon_{\text{mach}}) && \text{(local discrete incompressibility),} \\ r_{\text{port},i} &= \left| \sum_{\gamma \subset \partial \Omega_i} W_{i,\gamma}^{(0,n)} \right| && \text{(per-pore mass balance),} \\ r_{\text{flux}} &= \max_{\gamma} |W_{i,\gamma}^{(0,n)} + W_{j,\gamma}^{(0,n)}| && \text{(interface flux continuity),} \\ r_{\text{mom}} &= \max_{\gamma, \alpha, \beta} |W_{i,\gamma}^{(\alpha,\beta)} + W_{j,\gamma}^{(\alpha,\beta)}| && \text{(all nine moment continuity).} \end{aligned}$$

Here $W_{i,\gamma}^{(\alpha,\beta)}$ denotes the weighted velocity moment (equation (13)); only $W_{i,\gamma}^{(0,n)}$ is the volumetric flow rate. The implementation reports r_{mom} as `max.mass.residual` (computed from the full moment-residual vector and covering all nine modes); the naming is legacy and refers to the same quantity.

7 Relation to classic DD-PNM and scope of enrichment

Let $E_{P0} : \mathbb{R}^m \rightarrow \mathbb{R}^{9m}$ embed each classic coefficient into the constant-normal slot. The classic DD-PNM system is recovered by restricting **both trial and test spaces**:

$$\mathbf{S}_{P0} = E_{P0}^\top \mathbf{S} E_{P0}, \quad \mathbf{F}_{P0} = E_{P0}^\top \mathbf{F}.$$

An affine solution of $\mathbf{S} \boldsymbol{\lambda}^{\text{aff}} = \mathbf{F}$ generally does *not* have the classic solution as its constant-normal subvector, because the eight additional test-space constraints alter the Galerkin projection. Under the SPD hypotheses of Remark 1 this affine solution is unique.

The valid enrichment statement is conditional and reference-dependent. If a well-defined full interface-traction Schur system $\hat{\mathbf{S}} \boldsymbol{\xi} = \hat{\mathbf{F}}$ is introduced with the same local pressure spaces and sign conventions (the broken-pressure C-DD system of the companion mathematical properties document), then the affine system is its Galerkin restriction and the affine reduction error is no larger than the classic reduction error in the $\hat{\mathbf{S}}$ -energy seminorm. By the corresponding velocity-energy identity, the same monotonicity holds for the broken- H^1 velocity reduction error measured relative to the full C-DD solution. This argument does *not* imply monotonic improvement of velocity L^2 , pressure L^2 , or outlet-flux errors, nor of errors measured relative to the exact PDE solution.

No theorem in the present analysis establishes equality between that broken-pressure traction system and the monolithic continuous-pressure Taylor–Hood system. The “Exact FE-trace Schur” computation is a primal Schur reduction of the monolithic system. For its $O(10^{-12})$ agreement with the direct monolithic solve to hold, the two must operate on the **same assembled matrix**—identical pressure stabilisation, pressure constraints (pinning / zero-mean), boundary conditions, and gauge treatment. If the monolithic reference uses $\varepsilon = 0$ and the Schur construction uses $\varepsilon = 10^{-10}$, the matrices differ and the comparison is not an exact equivalence of a single discrete system.

8 Computational cost

A pore with m_i^u internal ports and m_i^k boundary ports requires $n_i = 9m_i^u + m_i^k$ primitive solves. Offline cost grows approximately $9\times$ relative to the one-mode-per-interface classic method. The global unknown count is $9m$; sparse factorization cost depends on graph fill and cannot be inferred from the unknown-count factor alone. For multi-boundary-condition studies the offline library is built once.

References

- [1] S. Sun, Z. Wang, L. Zhang, J. Zhao. A domain decomposition approach to pore-network modeling of porous media flow. *arXiv:2510.13429v2*, 2026.